



# AIRFLY INSIGHTS

- Overview
- Delay Analysis
- Cancellation Analysis
- Route Analysis
- Seasonal & Monthly Trends
- Airport-Level Performance Analysis
- Flight Distribution & Patterns
- Airline Performance Comparison
- Weather Impact Analysis
- Insights & Recommendations

~ Aravintakshan



# AIRFLY INSIGHTS | Overview

Year

2019

Month

All

Airline

All

484,551

Total Flights

12

Total Airlines

274

Total Airports

752.14

Average Distance

60.91

Average Arrival Delay

24079

Total Cancelled Flights

4.97

% Cancelled Flights



# AIRFLY INSIGHTS | Delay Analysis

Year

2019

Month

All

Airline

All

60.91

Avg Arrival Delay (mins)

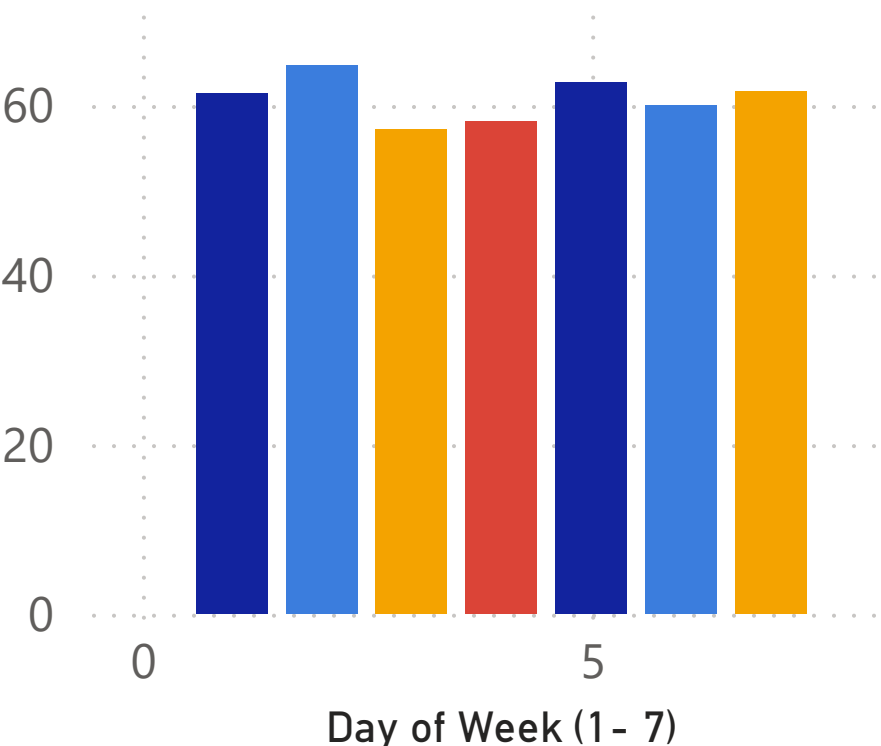
57.49

Avg Departure Delay (min)

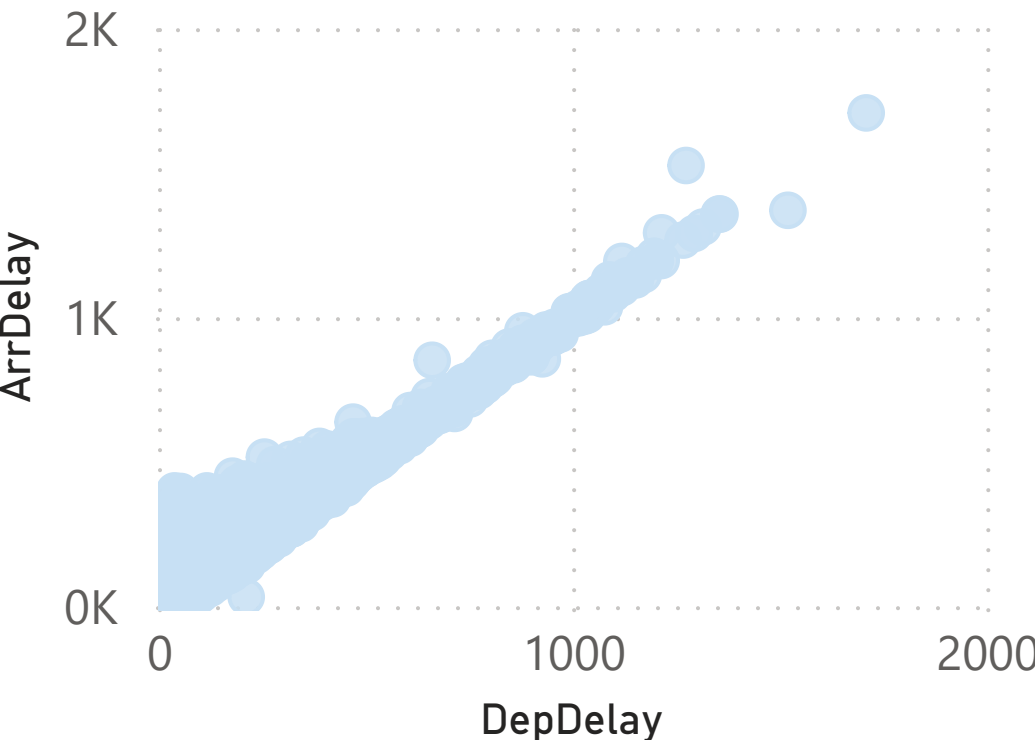
460472

Total Delayed Flights

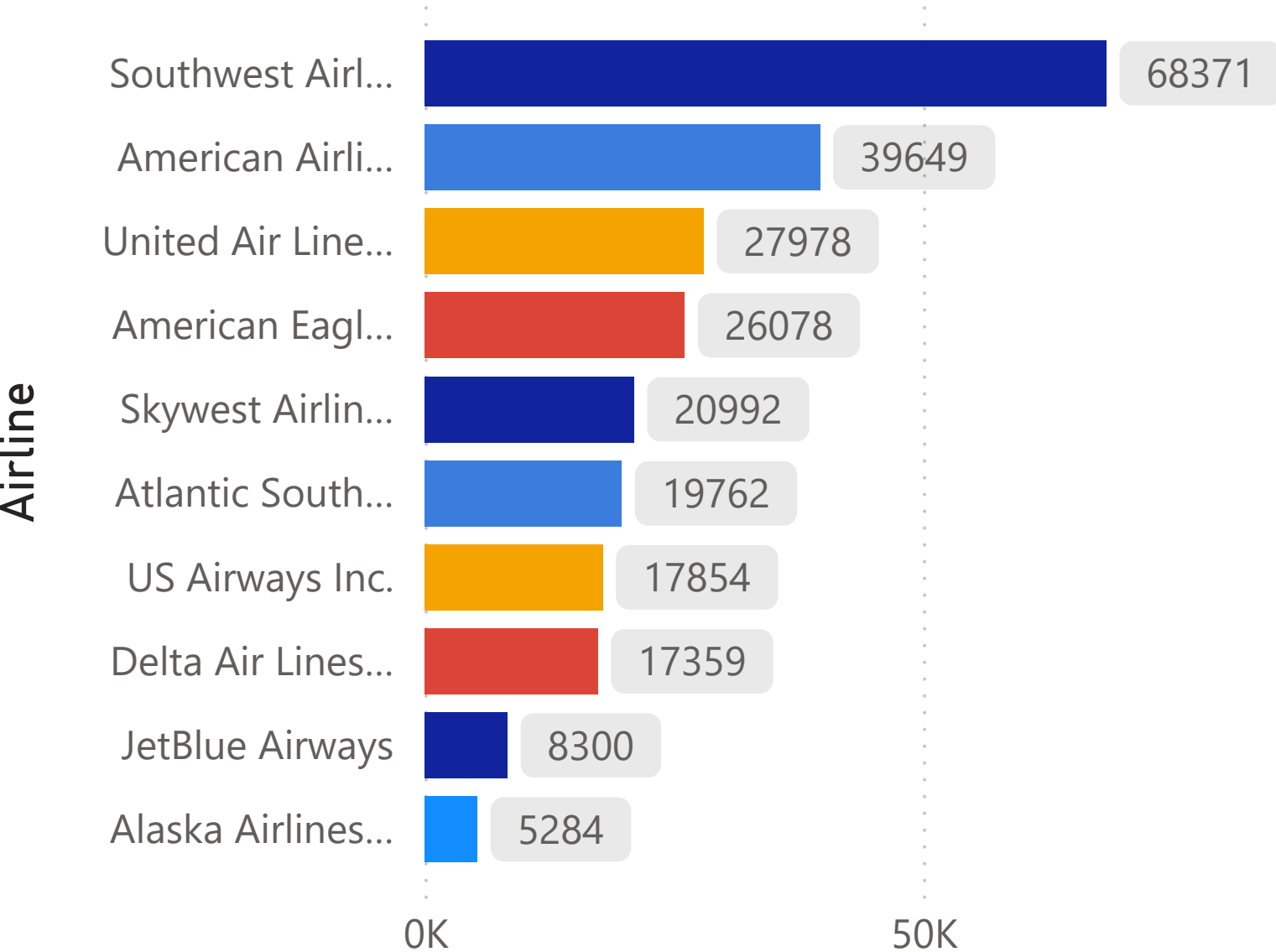
Delay by Day of Week



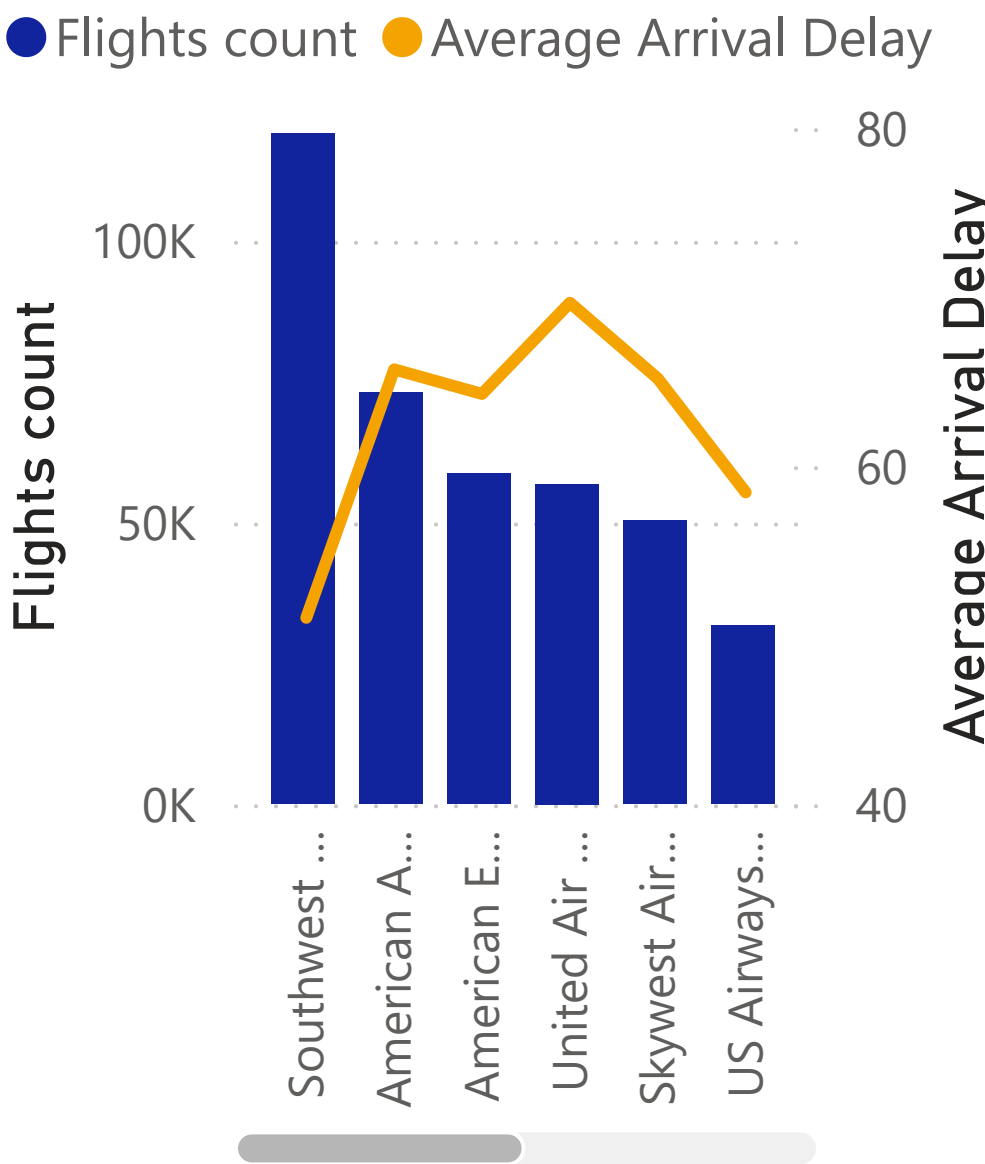
Departure Delay vs Arrival Delay



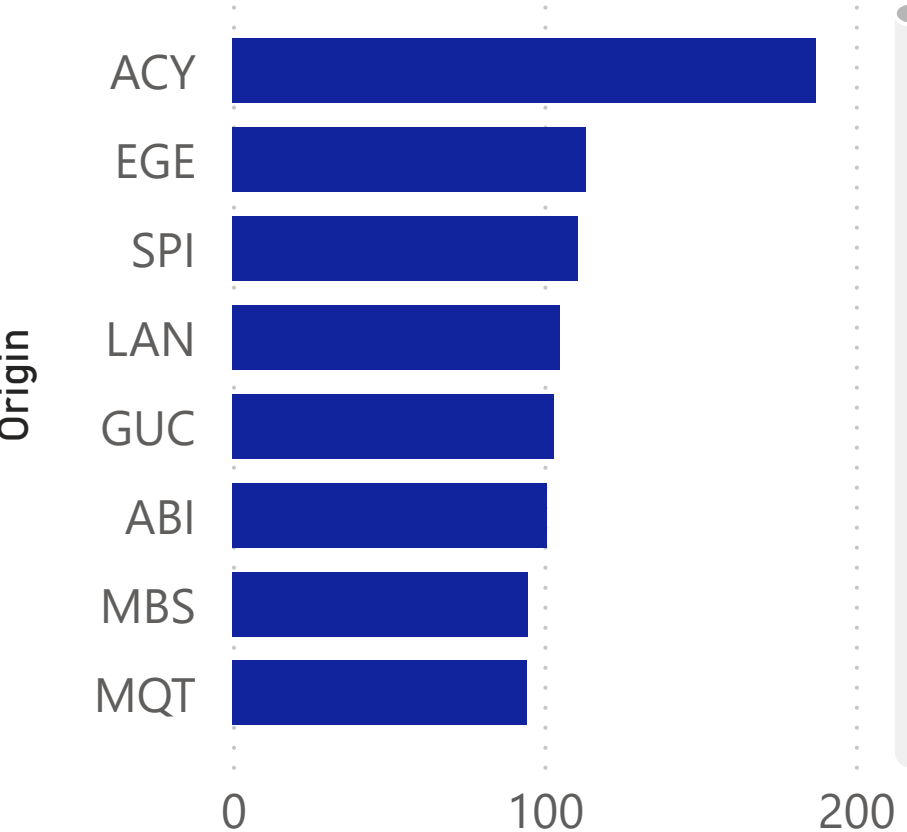
Top 10 Airlines by Carrier Delay Count



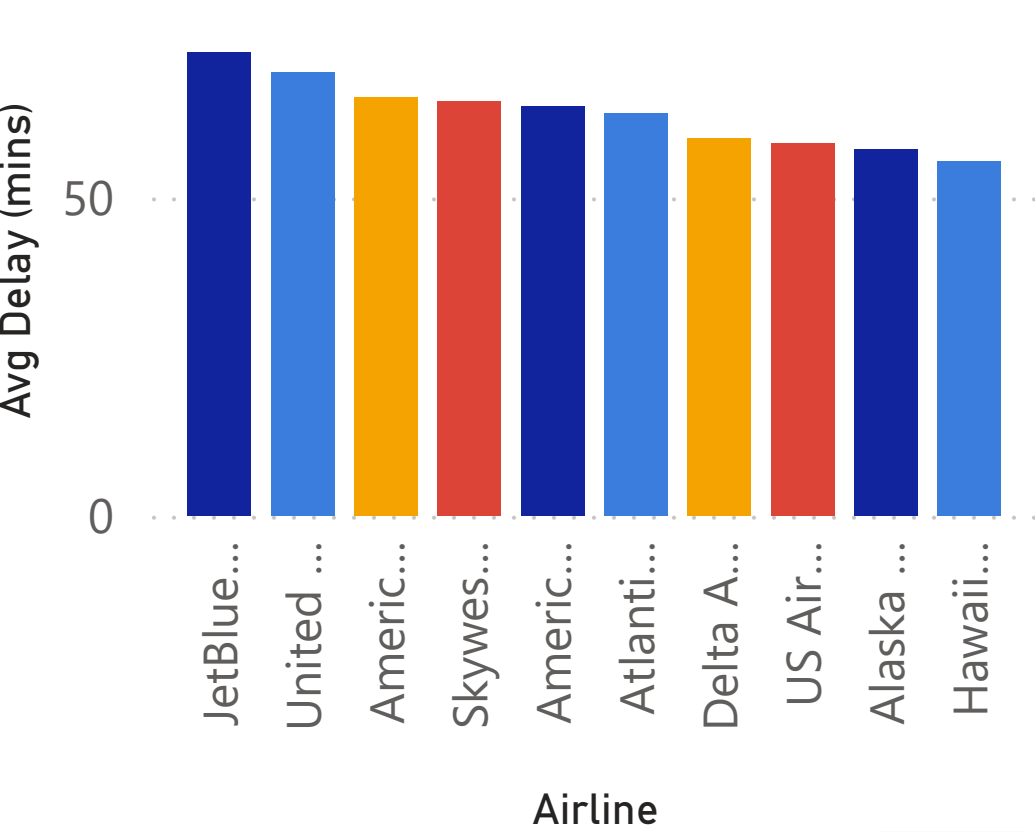
Flights vs Avg Delay



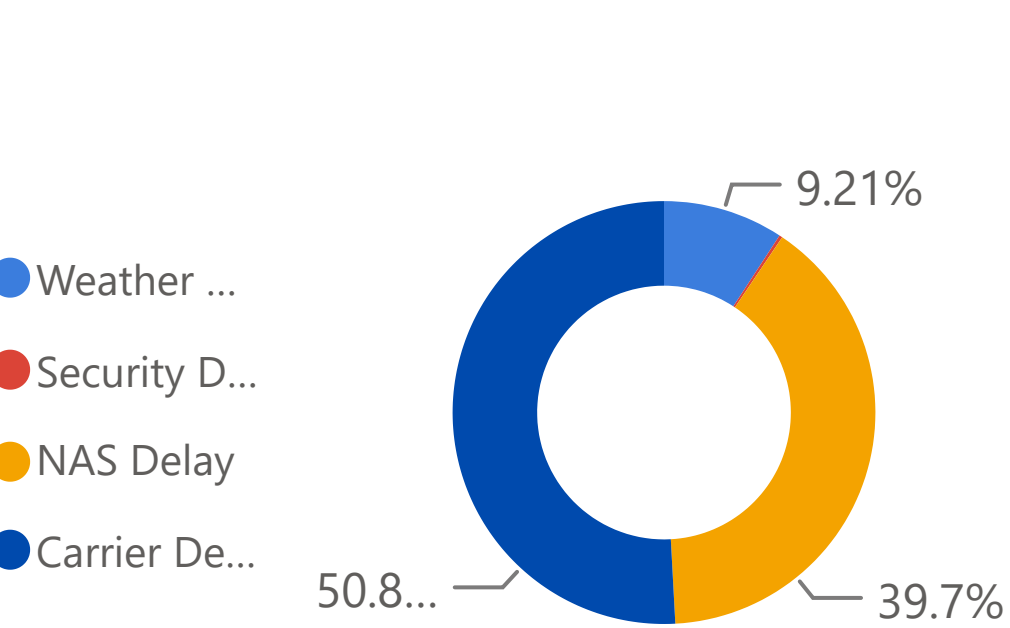
Average Delay by Airport



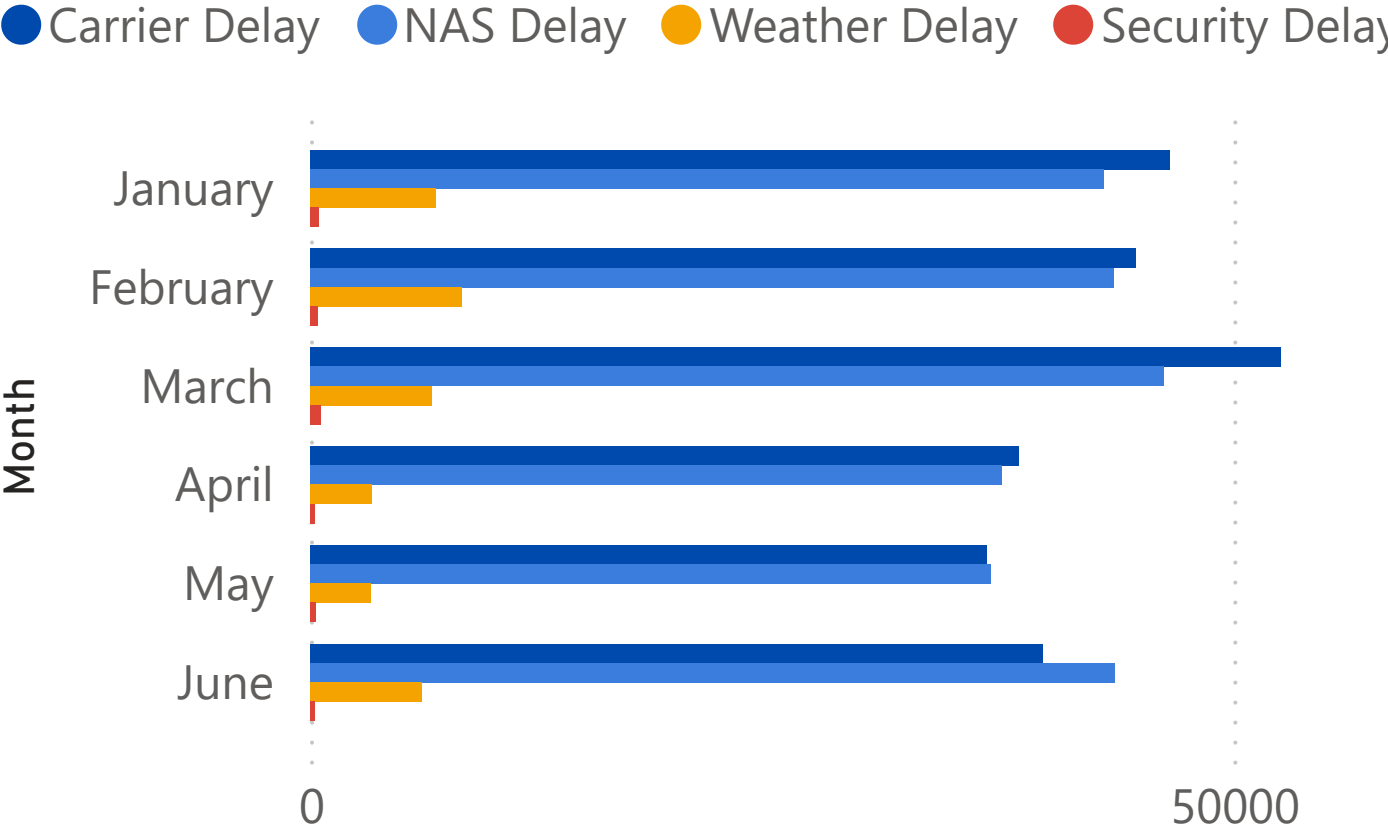
Average Delay by Airline

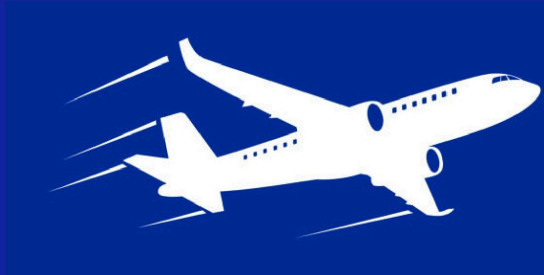


Total Delay by Cause



Delay Count by Reason (Monthly)





# AIRFLY INSIGHTS | Cancellation Analysis

Year

2019

Month

All

Airline

All

24079

Total Cancelled Flights

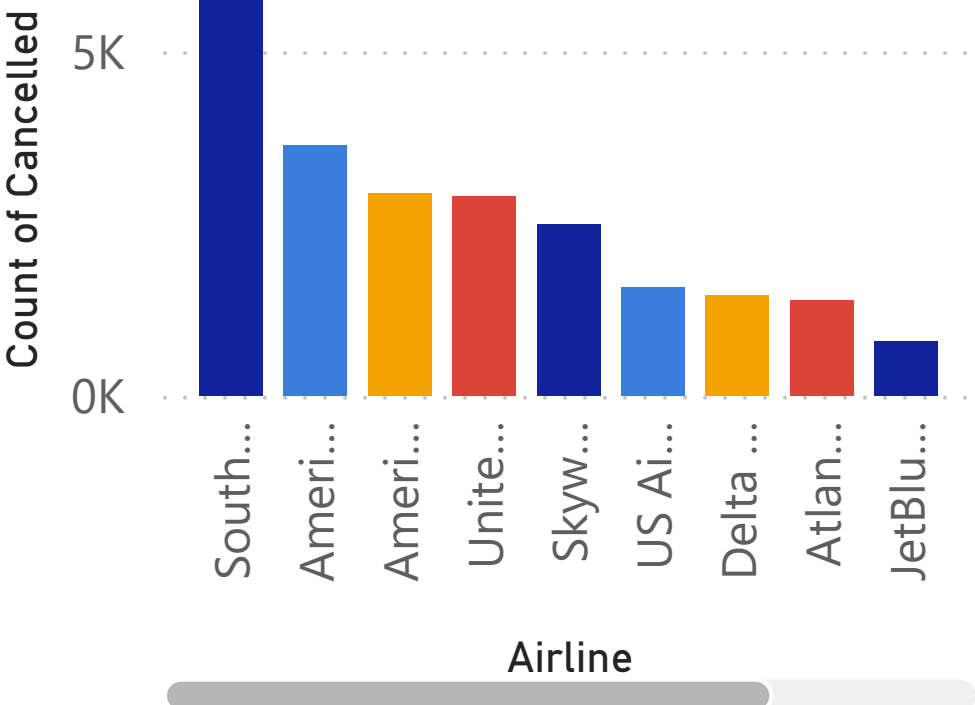
4.97

% Cancelled Flights

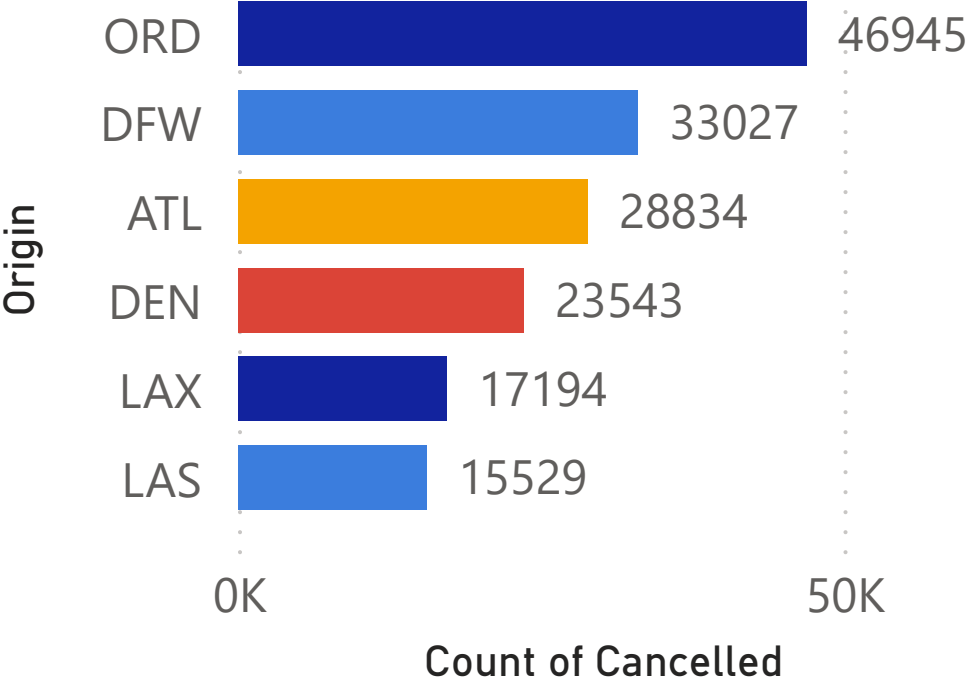
57.74

Avg Delay (Cancelled Flights)

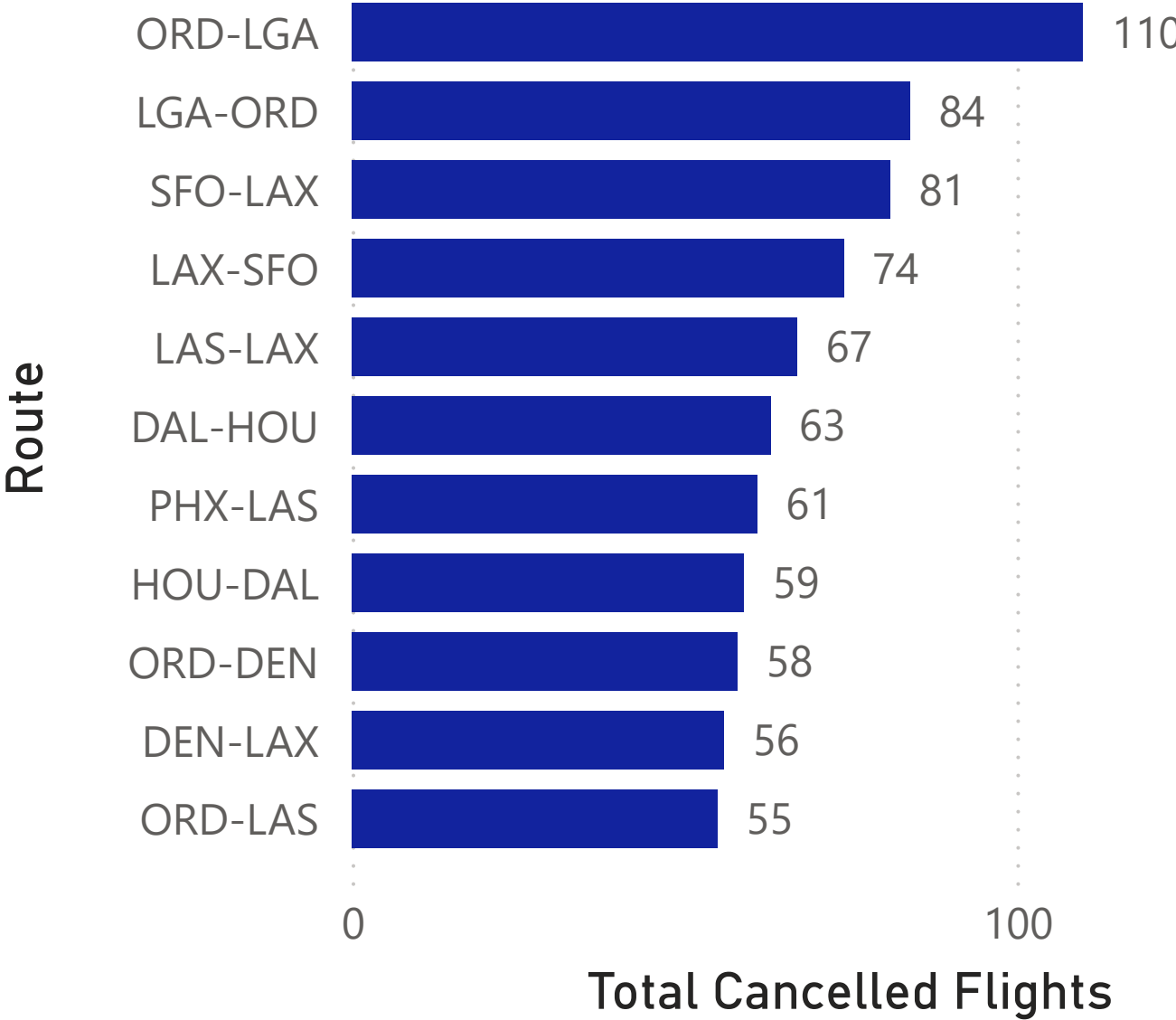
## Cancellations by Airline



## Cancellations by Airport

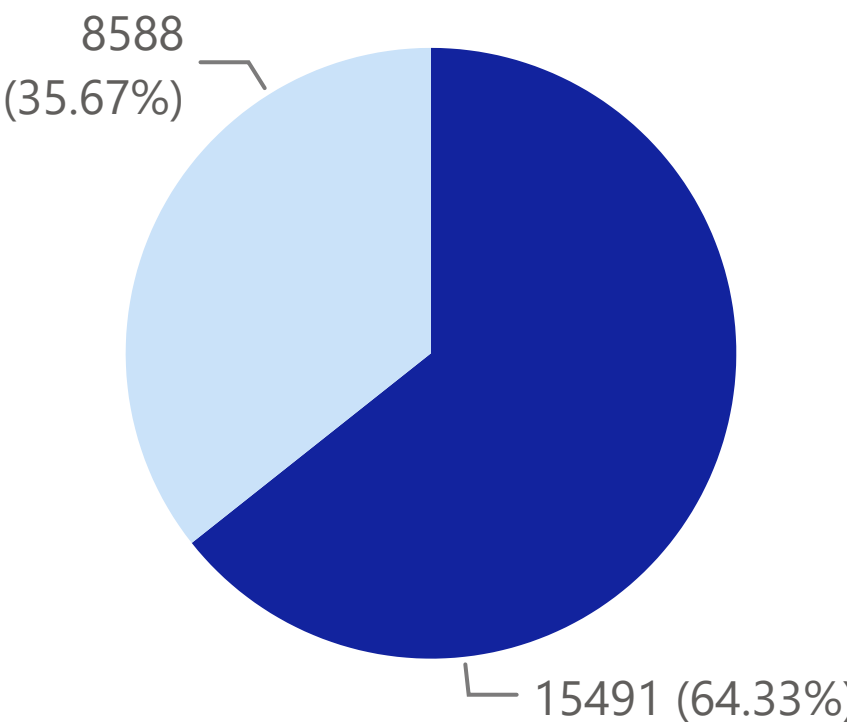


## Routes Affected by Cancellations

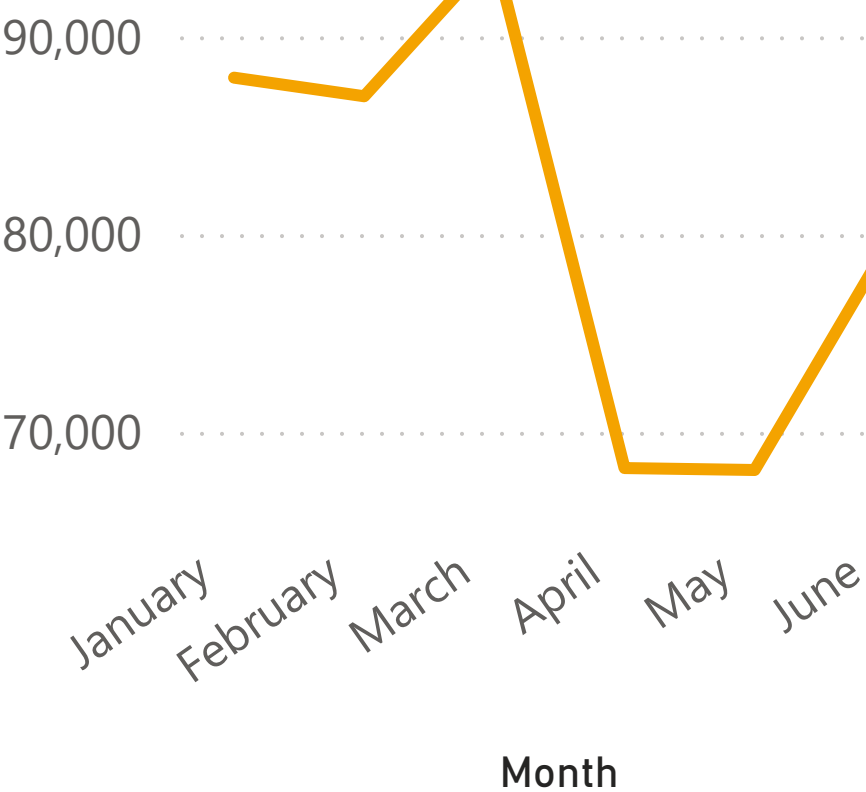


## Seasonal Cancellation

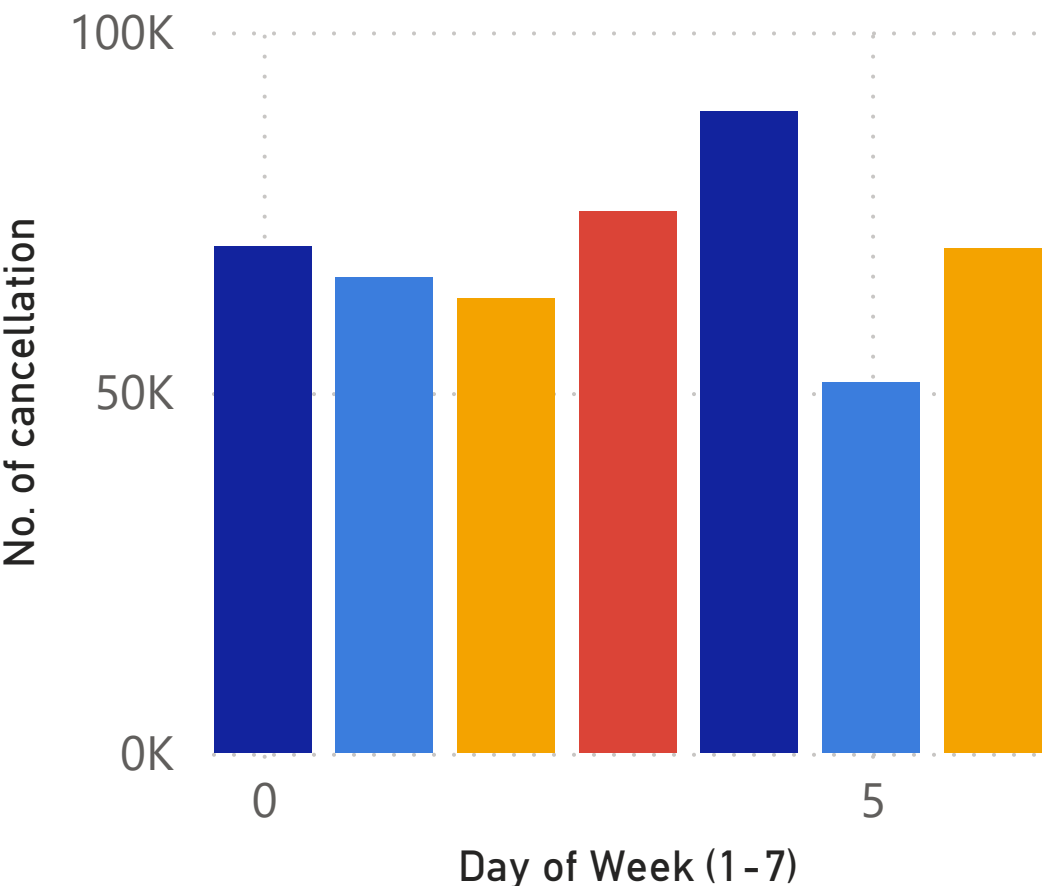
Season ● Non-Winter ● Winter



## Cancellations Over Time

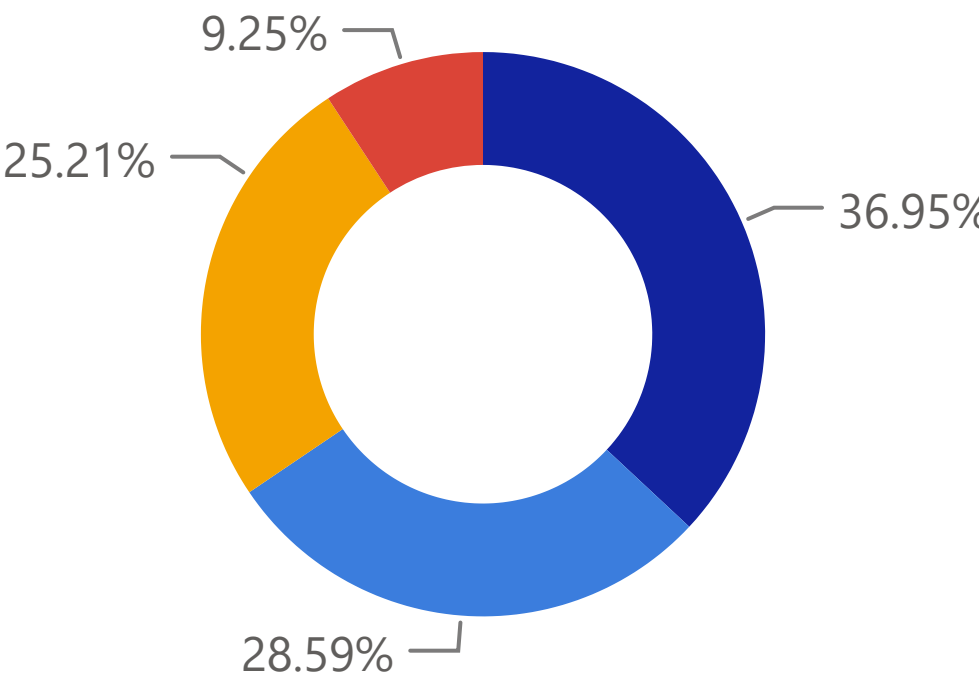


## Cancellations by Day of Week



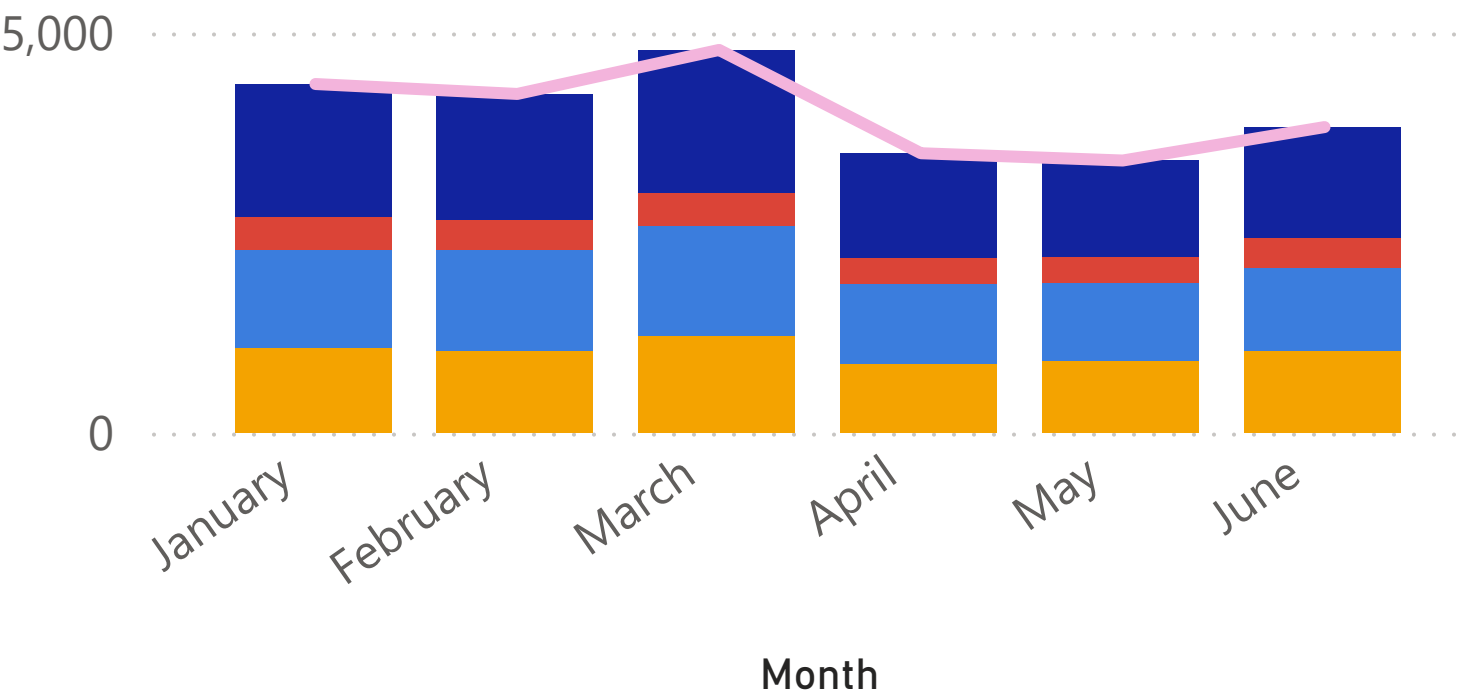
## Cancellations by Cause

Cancellat... ● Weather ● NAS ● Carrier ● Security



## Monthly Cancellation Trends by Cause

● Carrier ● NAS ● Security ● Weather ● Total Cancelled Flights





# AIRFLY INSIGHTS | Route Analysis

Year

Month

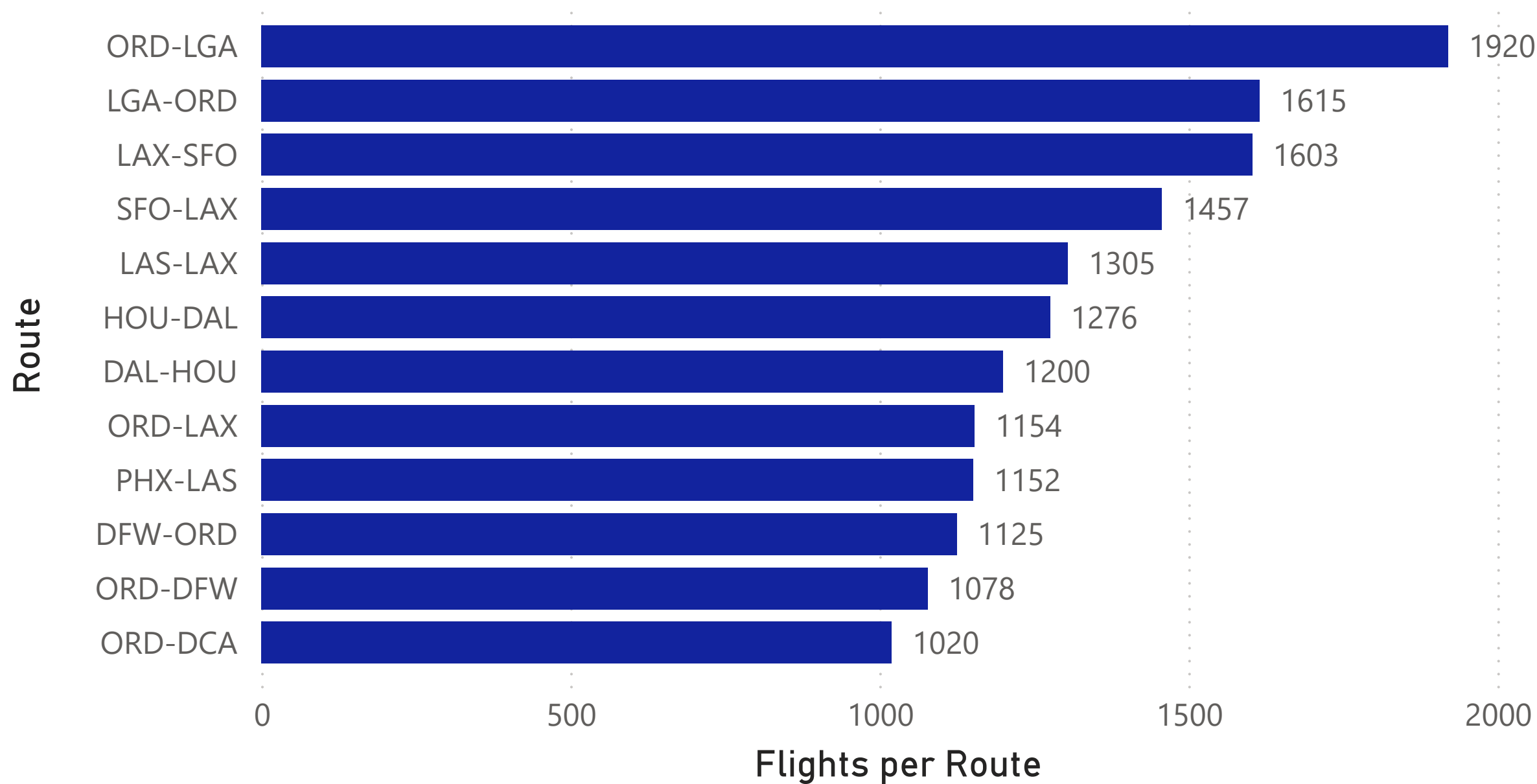
Airline

2019

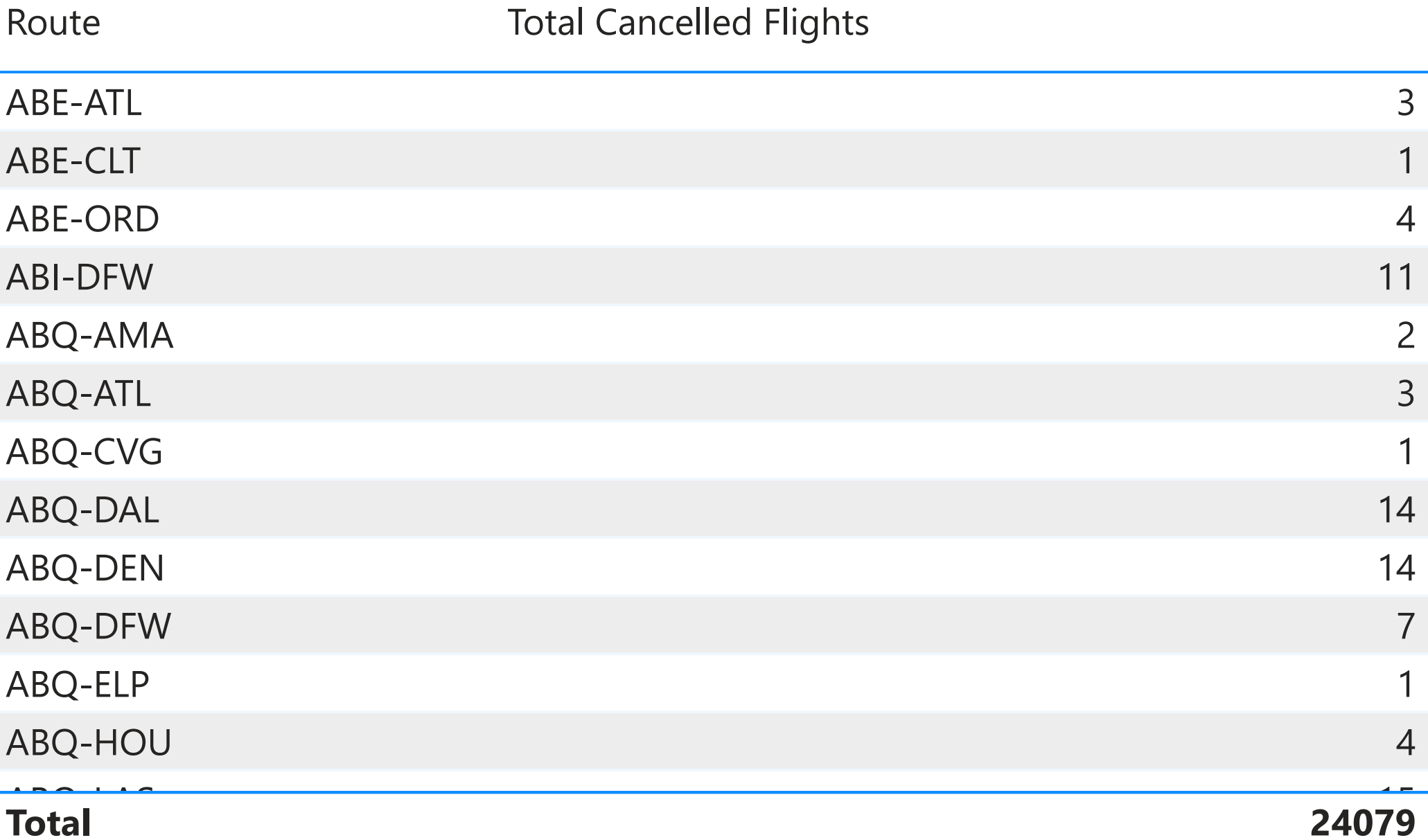
All

All

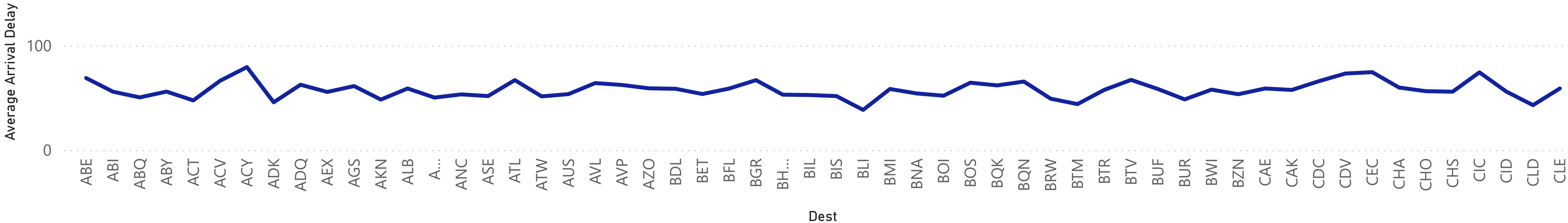
Top Origin-Destination



Routes Affected by Cancellations



Delay Analysis by Route





# AIRFLY INSIGHTS | Seasonal & Monthly Trends

Year

2019

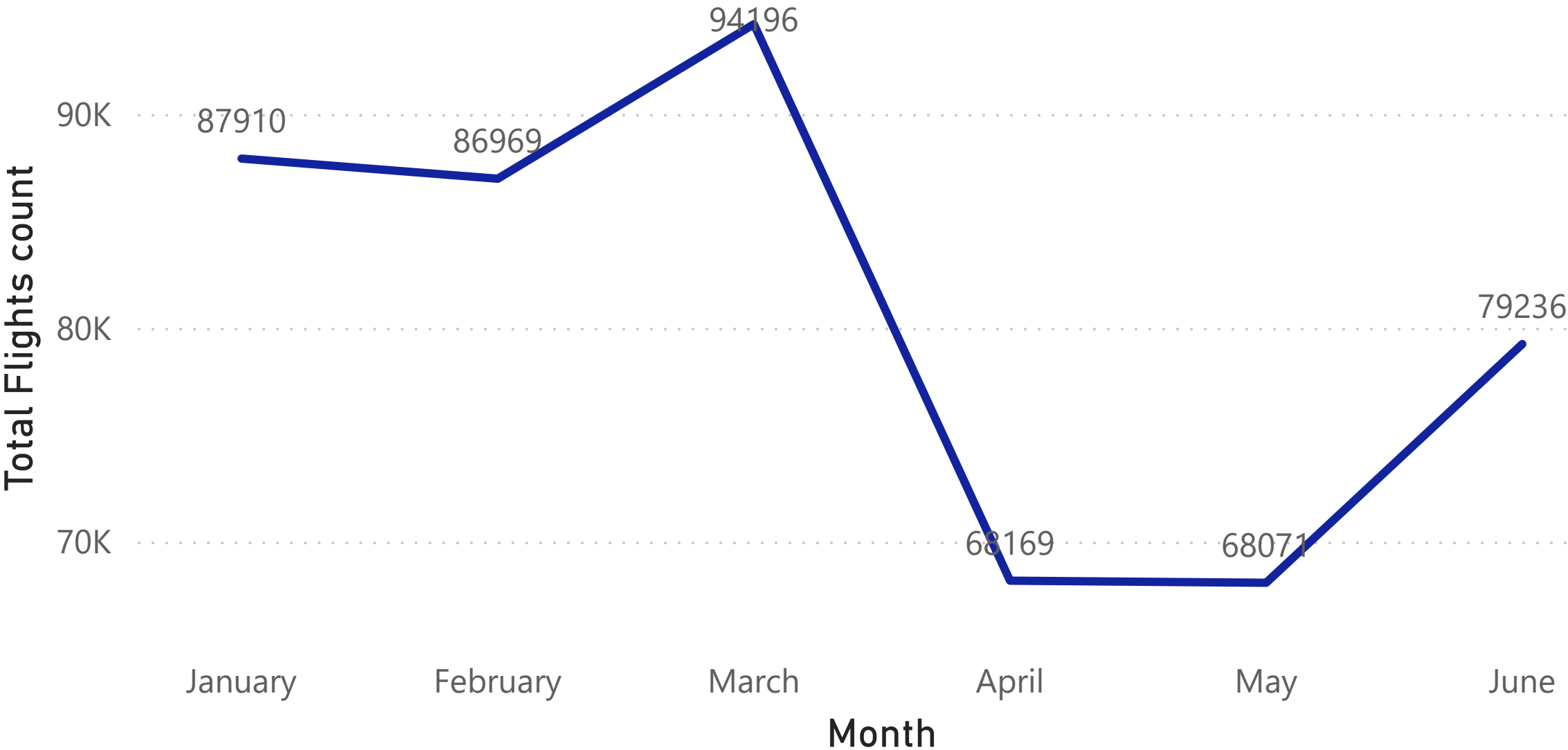
Month

All

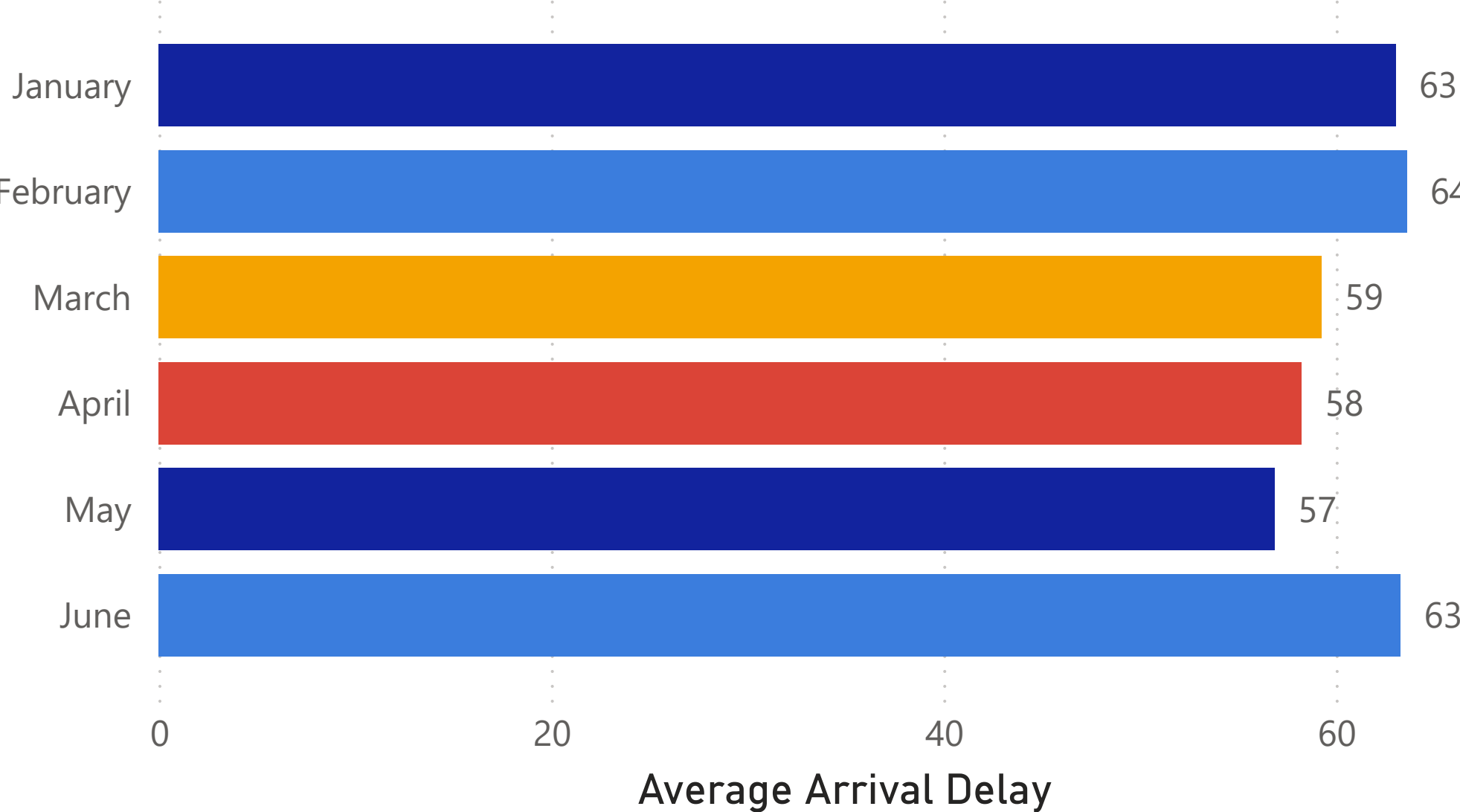
Airline

All

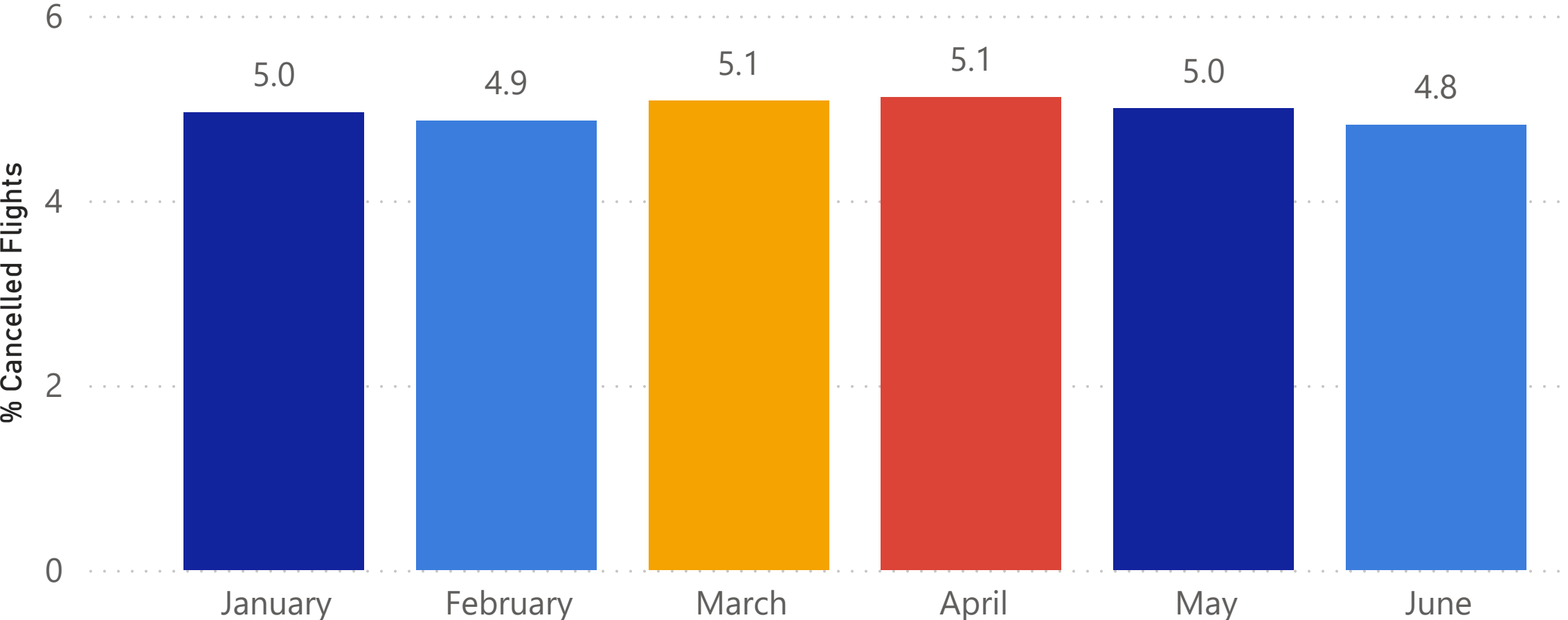
Month-wise Flight Counts



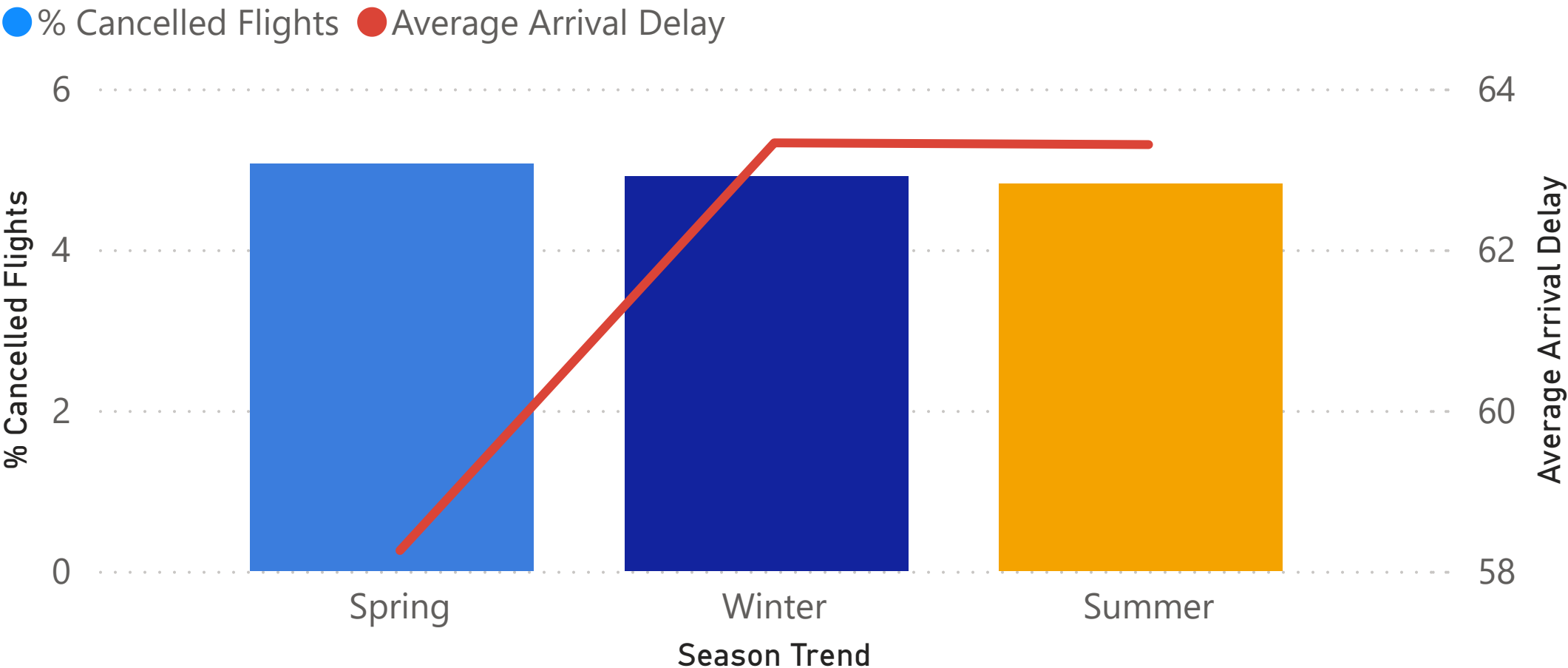
Monthly Delays



Monthly Cancellations



% Cancelled Flights and Average Arrival Delay by Season Trend





# AIRFLY INSIGHTS | Airport-Level Performance Analysis

Year

2019

Month

All

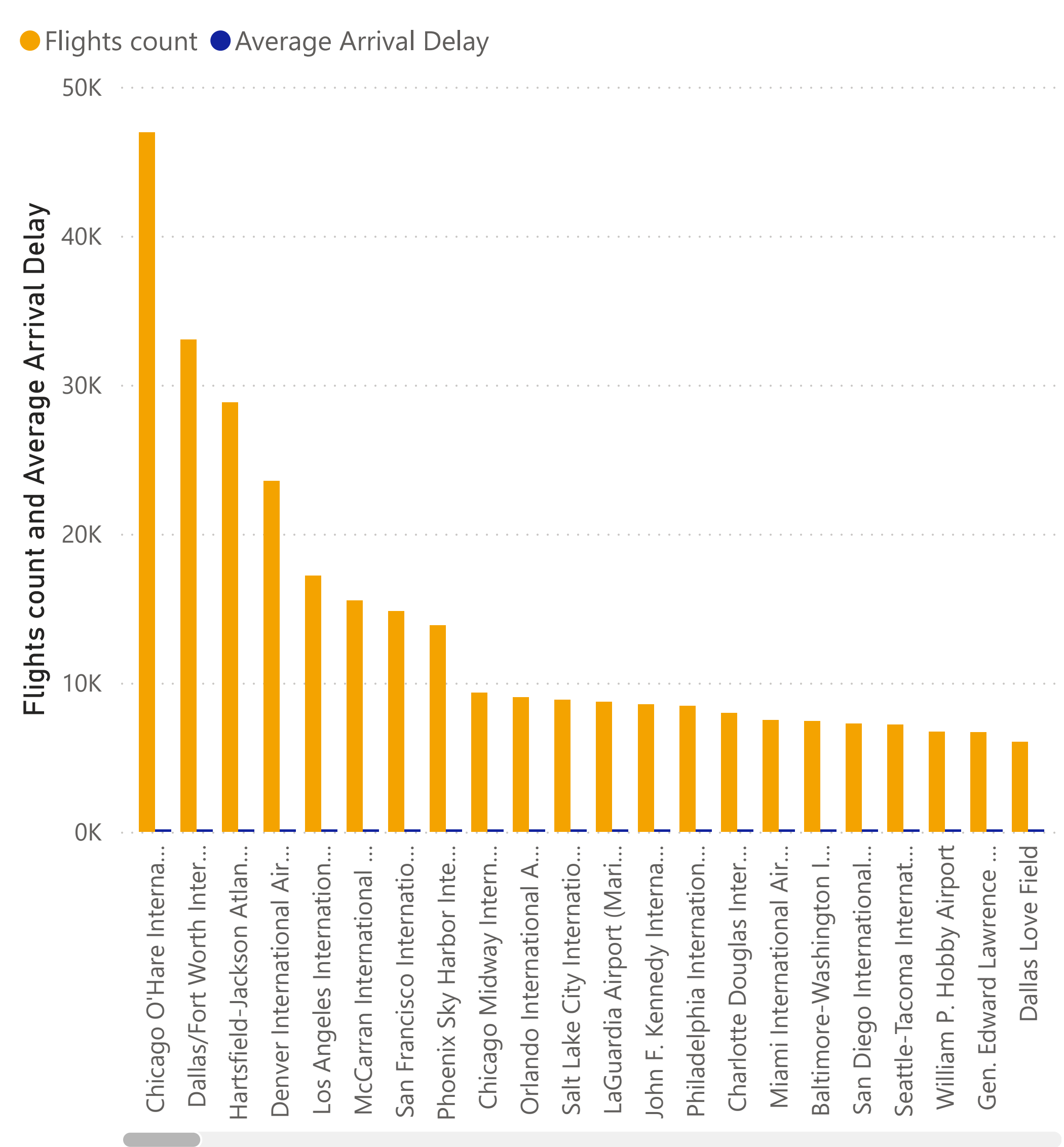
Airline

All

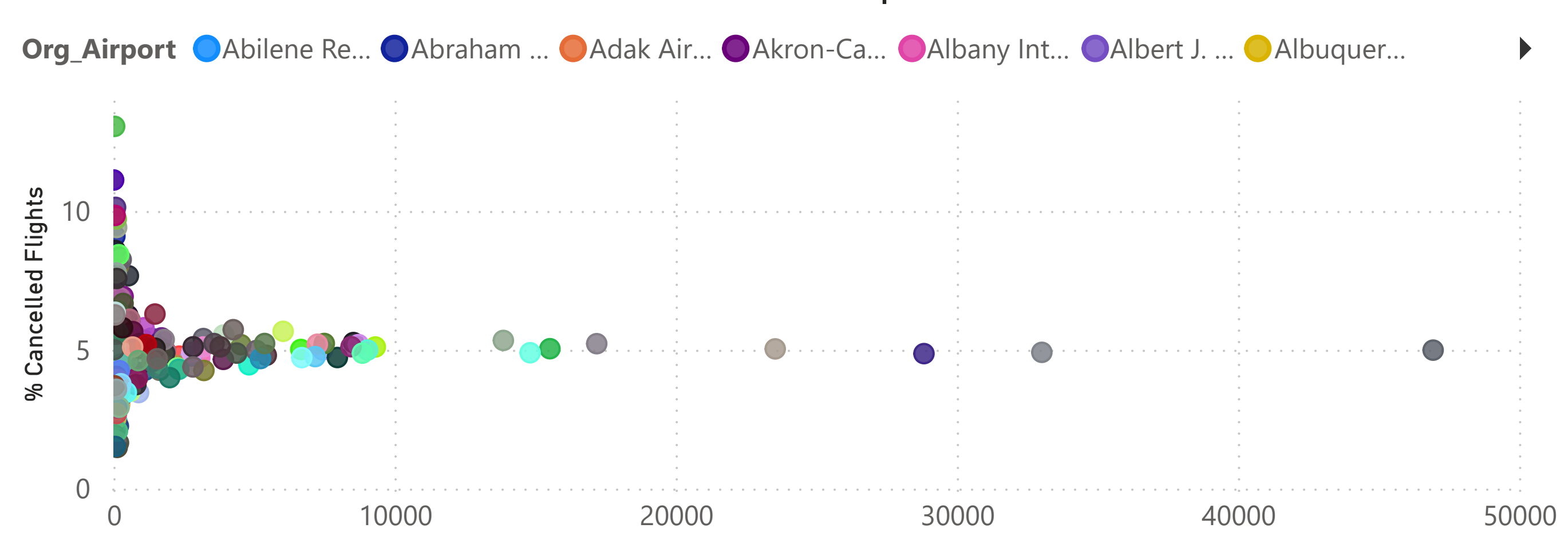
Airport with Avg delay

| Org_Airport                     | Dest_Airport                                       | Average Arrival Delay |
|---------------------------------|----------------------------------------------------|-----------------------|
| Abilene Regional Airport        | Dallas/Fort Worth International Airport            | 100.90                |
| Abraham Lincoln Capital Airport | Chicago O'Hare International Airport               | 111.09                |
| Adak Airport                    | Ted Stevens Anchorage International Airport        | 50.43                 |
| Akron-Canton Regional Airport   | Chicago O'Hare International Airport               | 117.63                |
| Akron-Canton Regional Airport   | Cincinnati/Northern Kentucky International Airport | 16.00                 |
| Akron-Canton Regional Airport   | Denver International Airport                       | 77.00                 |
| Akron-Canton Regional Airport   | Hartsfield-Jackson Atlanta International Airport   | 74.88                 |
| Albany International Airport    | Baltimore-Washington International Airport         | 57.01                 |
| Albany International Airport    | Charlotte Douglas International Airport            | 46.88                 |
| Albany International Airport    | Chicago Midway International Airport               | 58.48                 |
| Albany International Airport    | Chicago O'Hare International Airport               | 70.87                 |
| Total                           |                                                    | 60.91                 |

Flights vs Avg Delay



Cancellation % vs Airport Traffic





# AIRFLY INSIGHTS | Flight Distribution & Patterns

Year

2019

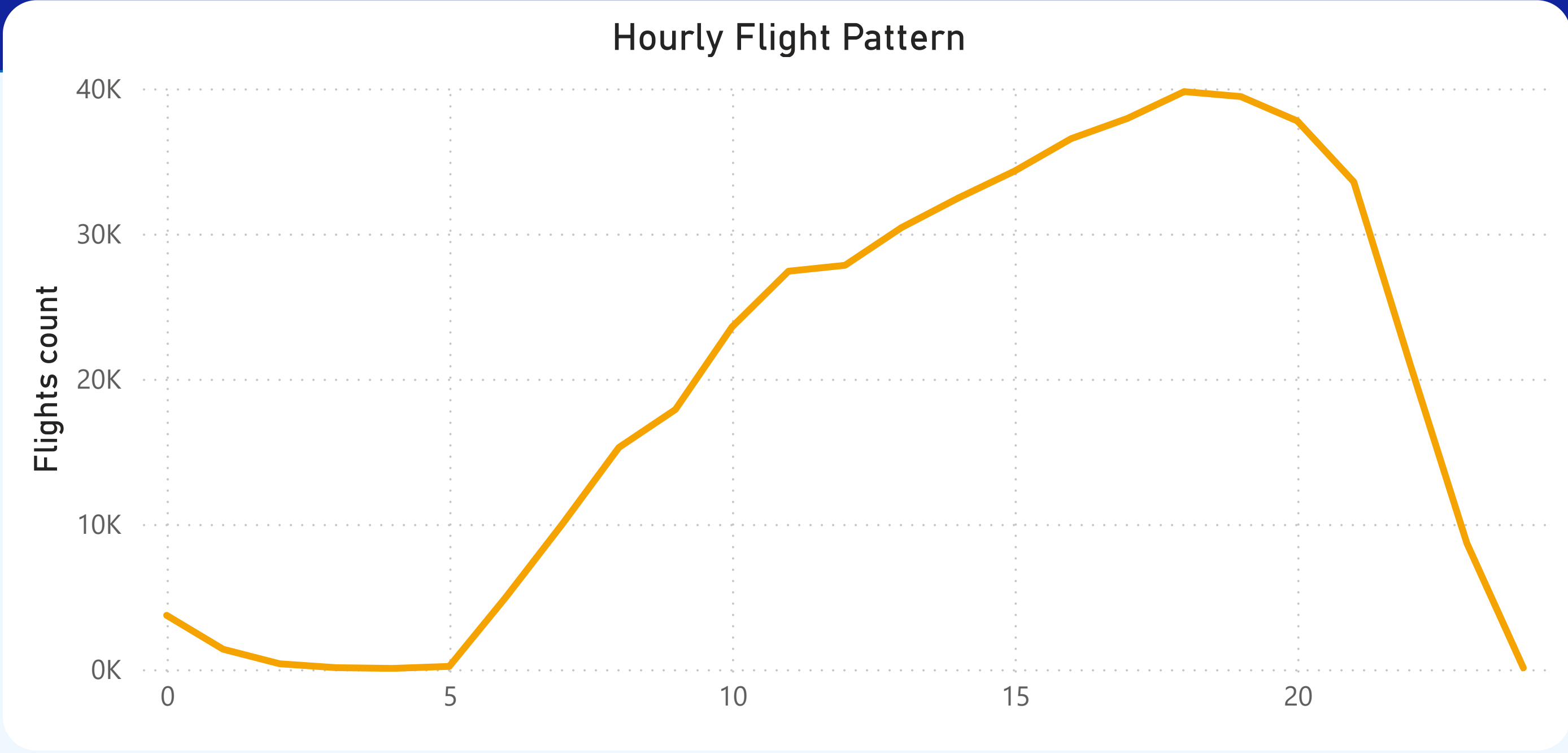
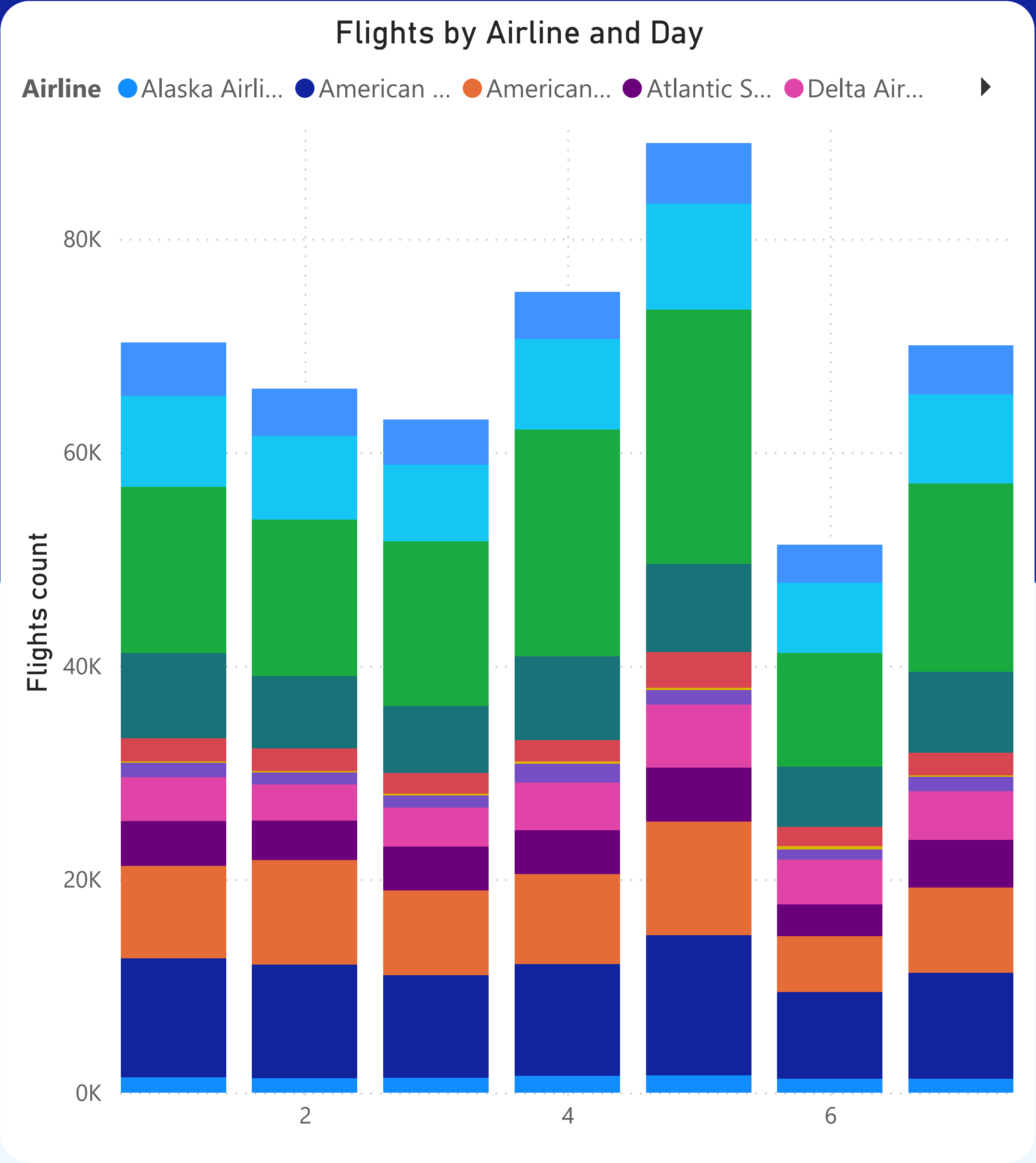
Month

All

Airline

All

| Flights by Day of Week & DepHour |      |      |     |     |    |     |      |      |       |       |       |       |       |       |
|----------------------------------|------|------|-----|-----|----|-----|------|------|-------|-------|-------|-------|-------|-------|
| DayOfWeek                        | 0    | 1    | 2   | 3   | 4  | 5   | 6    | 7    | 8     | 9     | 10    | 11    | 12    | 13    |
| 1                                | 426  | 170  | 40  | 15  | 4  | 30  | 763  | 1632 | 2431  | 2796  | 3775  | 4208  | 4363  | 4503  |
| 2                                | 453  | 164  | 47  | 7   | 8  | 21  | 682  | 1509 | 2182  | 2642  | 3273  | 3877  | 3796  | 4242  |
| 3                                | 434  | 192  | 57  | 16  | 7  | 26  | 657  | 1415 | 2091  | 2555  | 3164  | 3612  | 3592  | 3922  |
| 4                                | 601  | 190  | 49  | 13  | 6  | 29  | 709  | 1321 | 2192  | 2502  | 3436  | 3912  | 3892  | 4415  |
| 5                                | 798  | 289  | 60  | 17  | 10 | 26  | 856  | 1753 | 2601  | 3067  | 3995  | 4806  | 4920  | 5410  |
| 6                                | 344  | 156  | 59  | 18  | 12 | 31  | 724  | 1312 | 2039  | 2317  | 3011  | 3453  | 3522  | 3676  |
| 7                                | 653  | 212  | 60  | 17  | 7  | 29  | 563  | 1052 | 1731  | 2001  | 2907  | 3537  | 3727  | 4242  |
| Total                            | 3709 | 1373 | 372 | 103 | 54 | 192 | 4954 | 9994 | 15267 | 17880 | 23561 | 27405 | 27812 | 30410 |





# AIRFLY INSIGHTS | Airline Performance Comparison

Year

2019

Month

All

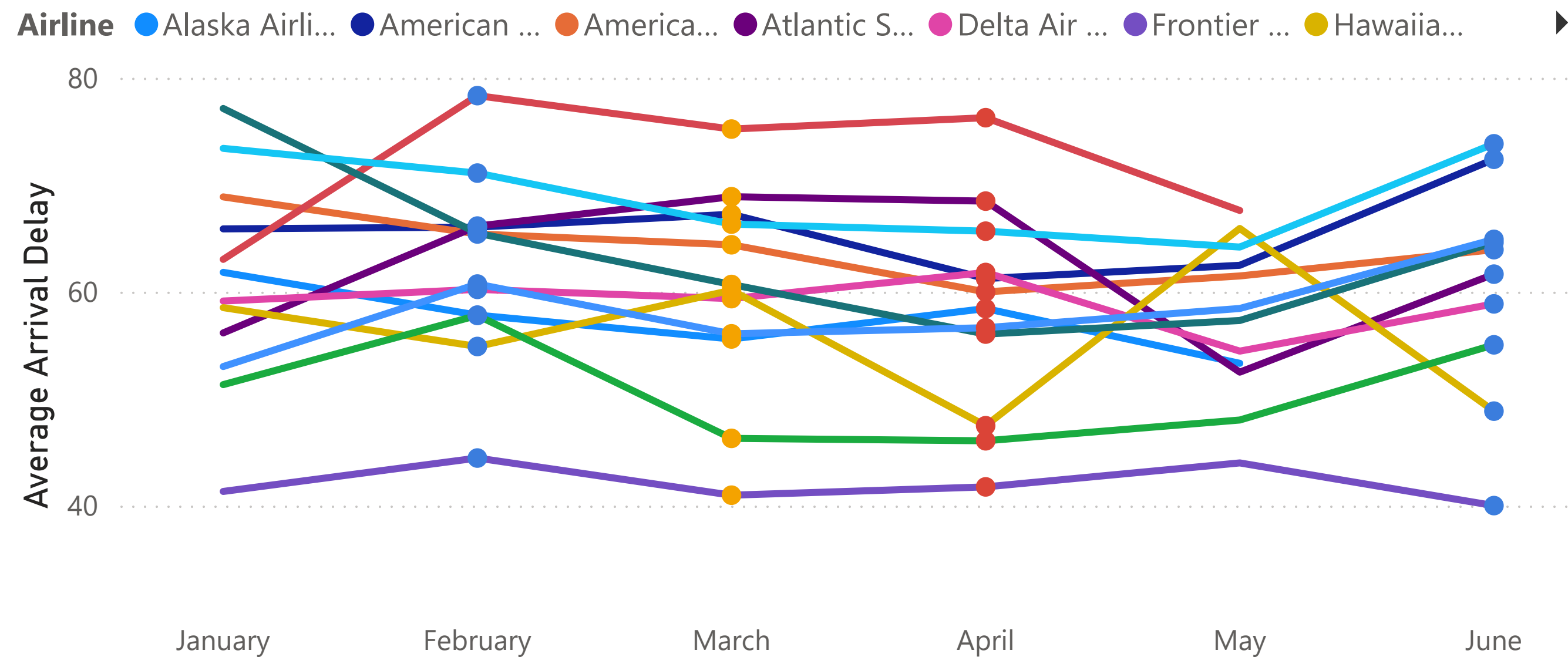
Airline

All

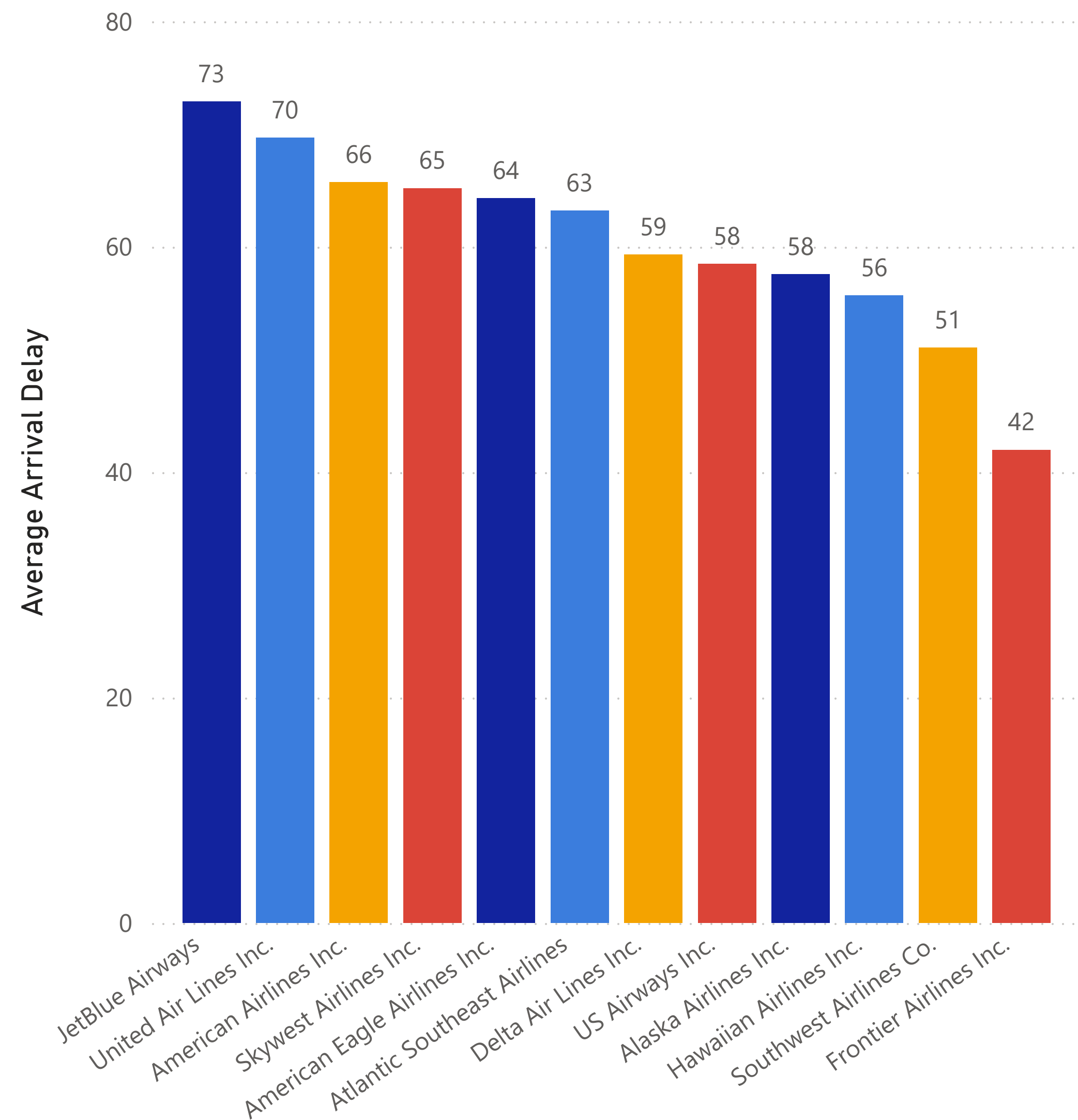
Monthly Cancellation % by Airline

| Airline                      | January | February | March | April | May  | June | Total |
|------------------------------|---------|----------|-------|-------|------|------|-------|
| Alaska Airlines Inc.         | 4.46    | 5.65     | 5.64  | 5.35  | 4.83 |      | 5.16  |
| American Airlines Inc.       | 4.95    | 5.10     | 5.08  | 4.91  | 5.06 | 4.63 | 4.97  |
| American Eagle Airlines Inc. | 4.83    | 4.86     | 4.86  | 5.44  | 5.27 | 4.91 | 5.00  |
| Atlantic Southeast Airlines  | 4.94    | 4.50     | 4.59  | 5.38  | 4.89 | 4.90 | 4.84  |
| Delta Air Lines Inc.         | 4.56    | 4.87     | 5.01  | 4.83  | 4.84 | 4.50 | 4.81  |
| Frontier Airlines Inc.       | 4.98    | 3.81     | 5.46  | 5.67  | 4.43 | 5.09 | 4.90  |
| Hawaiian Airlines Inc.       | 7.04    | 5.66     | 8.86  | 5.61  | 3.61 | 2.33 | 4.93  |
| JetBlue Airways              | 5.06    | 5.20     | 5.59  | 4.67  | 4.94 |      | 5.14  |
| Total                        | 4.96    | 4.87     | 5.08  | 5.12  | 4.99 | 4.82 | 4.97  |

Monthly Delay Trend by Airline



Average Delay per Airline





# AIRFLY INSIGHTS | Weather Impact Analysis

Year

2019

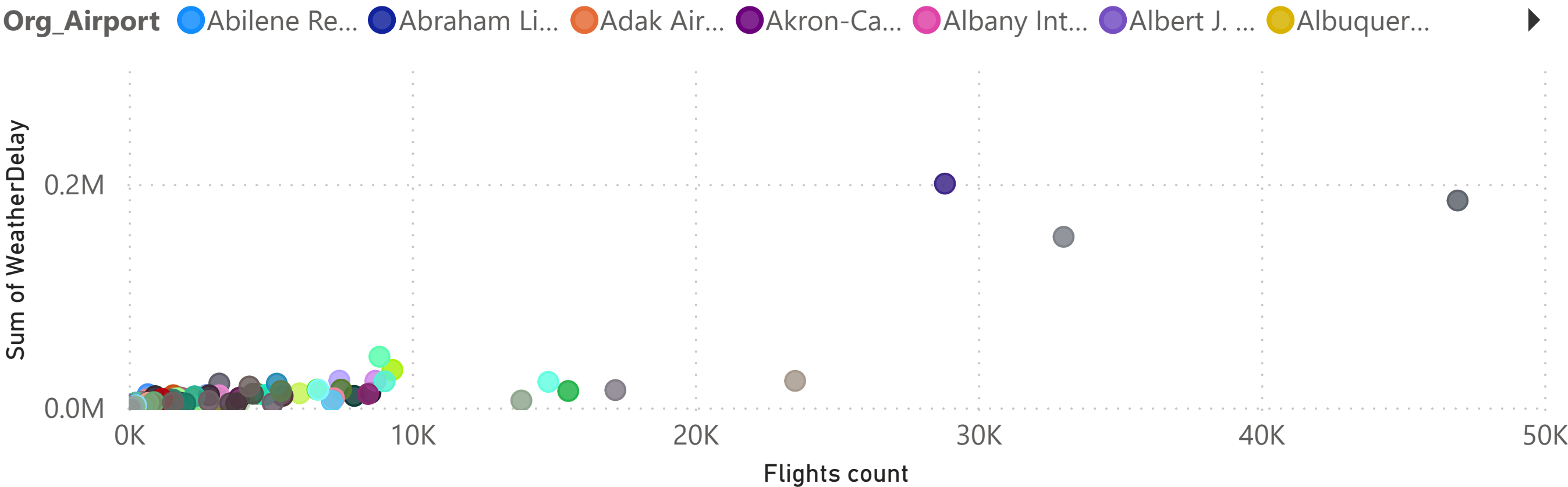
Month

All

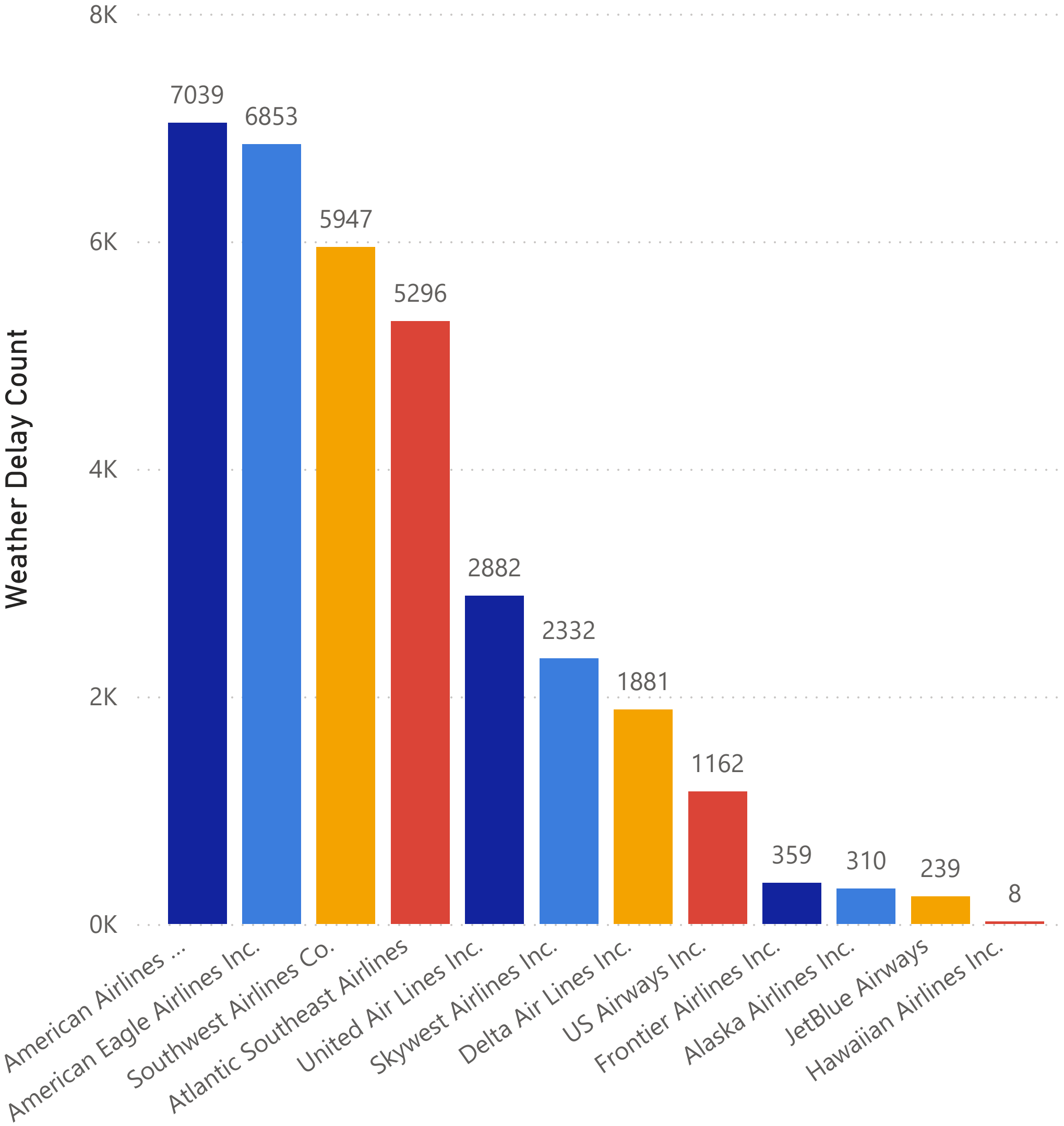
Airline

All

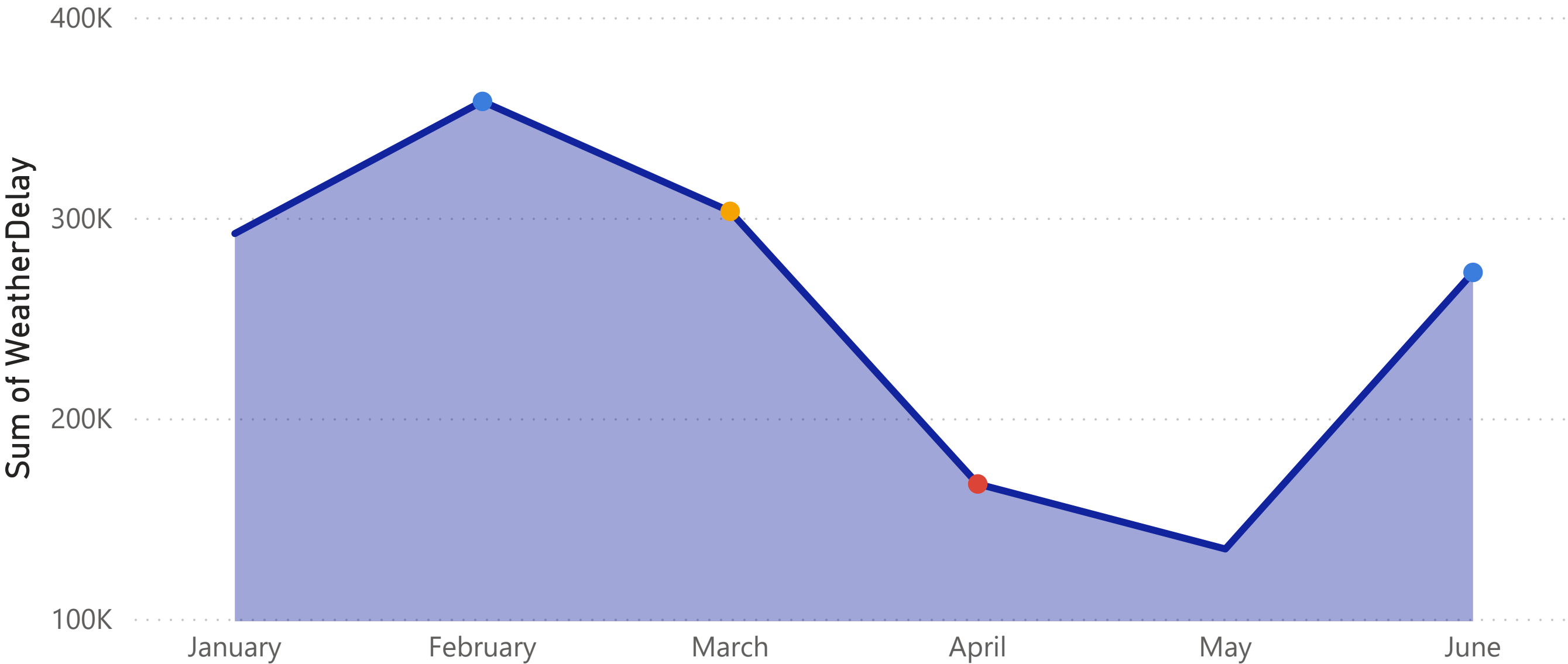
Weather Delay vs Total Flights



Average Delay per Airline



Weather-Related Delays Over Months





# AIRFLY INSIGHTS | Insights & Recommendations

**Carrier delays** contribute more to overall delays compared to NAS, weather, or security delays.

**ORD (Chicago O'Hare)** has the **maximum cancelled flights**, especially on the **ORD → LGA** route.

**Southwest Airlines** experiences the **highest carrier delays** and also records the **maximum number of cancellations**.

**March** shows the **highest carrier delays**, indicating seasonal operational challenges.

**ORD - LGA** has top origin to destination though it has most cancelled flights.

**WYS Airport** has very least delayed and cancelled flights.

**American Airlines Inc.** has highest weather-related delays.

**Frontier Airline Inc.** has least above cancellation and average delay.

**Hawaiian Airlines** has least weather impact & lowest cancellations.

**February** has highest average delayed flights and in **March** has highest cancelled flights.

**Hawaiian Airlines** has highest airline performance.

